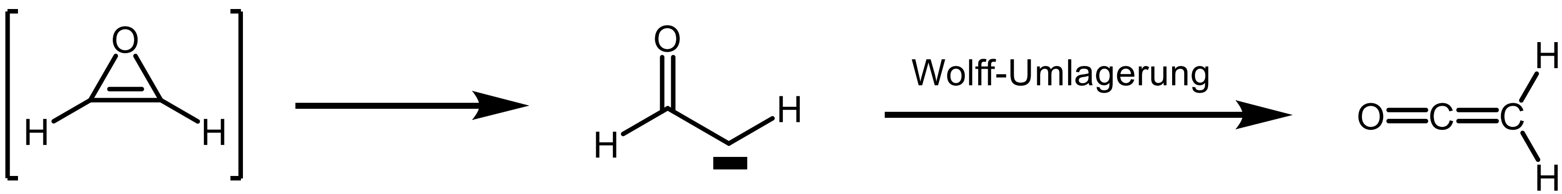
3. Kleine Ringe

3.1. Dreiring-Heterocyclen

Ungesättigte (Oxiren, Thiiren, 2*H*-Azirin, 3*H*-Diazirin) und gesättigte Systeme (Oxiran, Dioxiran, Aziridin)

3.1.1. Ungesättigte Dreiringheterocyclen

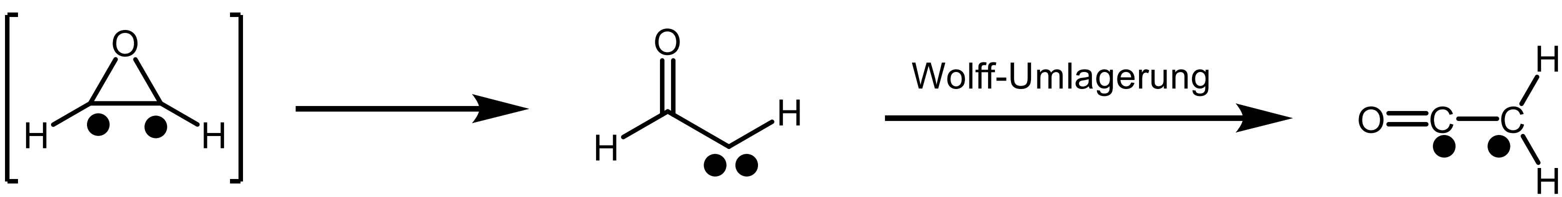
3.1.1.1. Oxiren



4π-Antiaromat Singulett-Oxiren (unbekannt) 45 kcal/mol

Singulett-Formylcarben 44 kcal/mol

Singulett-Keten -34 kcal/mol



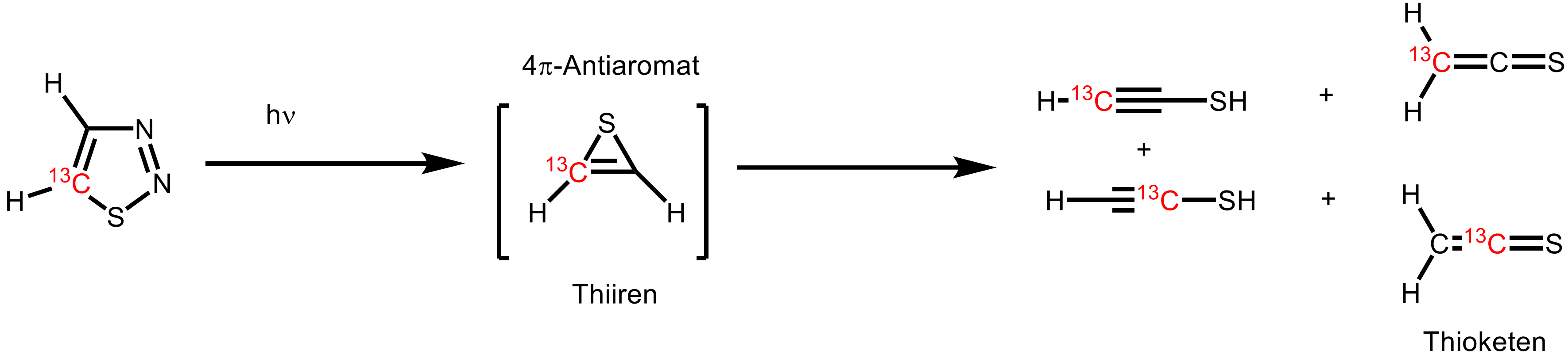
Triplett-Oxiren (unbekannt) 69 kcal/mol

Triplett-Formylcarben 41 kcal/mol

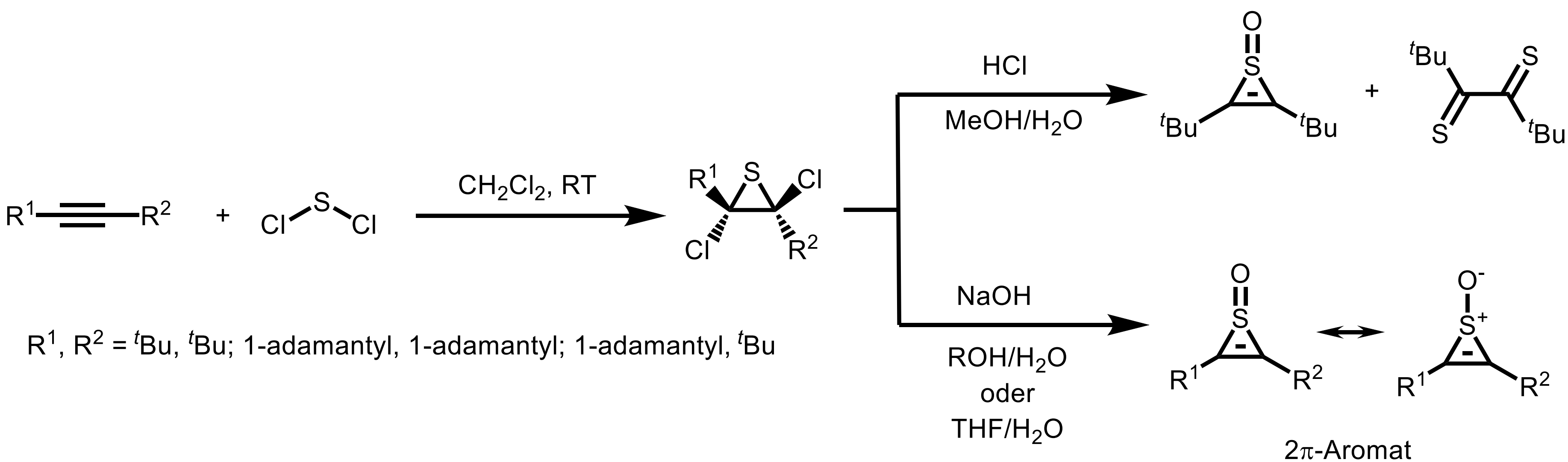
Triplett-Keten 20 kcal/mol

3.1.1.2. Thiiren

Matrixisolation, Isotopen: A. Krantz, J. Laureni, *J. Am. Chem. Soc.* **1981**, *103*, 486.

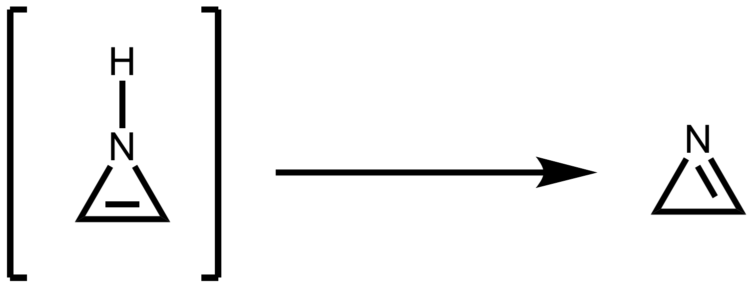


Thiiren-S-oxid: J. Nakayama, K. Takahashi, Y. Sugihara, A. Ishii, *Tetrahedron Lett.* **2001**, *42*, 4017.



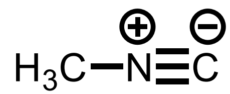
R^1, R^2 = ^{t}Bu, ^{t}Bu; 1-adamantyl, 1-adamantyl; 1-adamantyl, ^{t}Bu

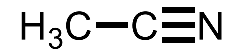
3.1.1.3. 2*H*-Azirin



1*H*-Azirin

2*H*-Azirin





Stabilomer

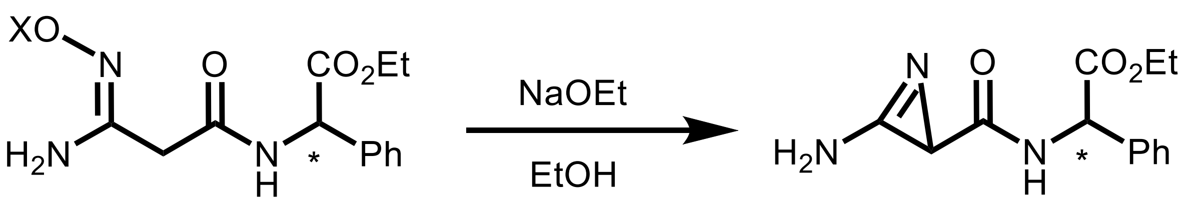
Das antiaromatische 4π-System kann ohne Ringöffnung umgangen werden:

Abwinklung von 1*H* oder besser durch Tautomerisierung zum 2*H*-Isomer

Spannungsenthalpie von 2*H*-Azirin ~190 $kJmol^{-1}$ (vgl. Cyclopropen ~ 230 $kJmol^{-1}$)

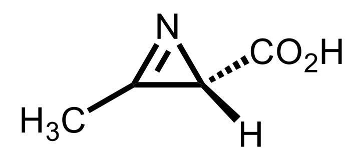
Synthesen

a) Modifizierte Neber-Reaktion



CH_3SO_2Cl X = H → X = SO_2Me

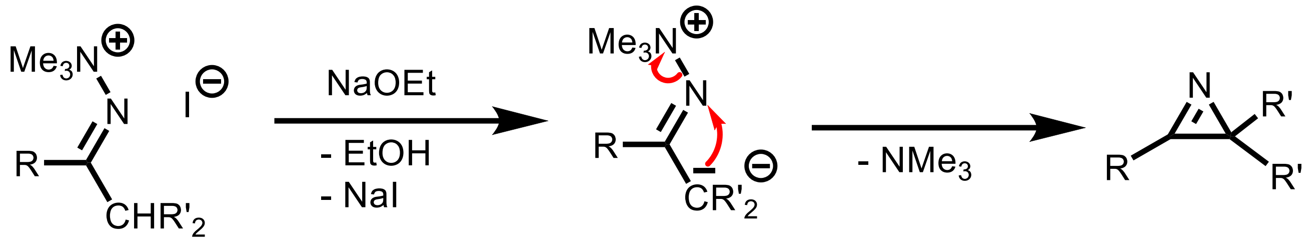
de > 90 %



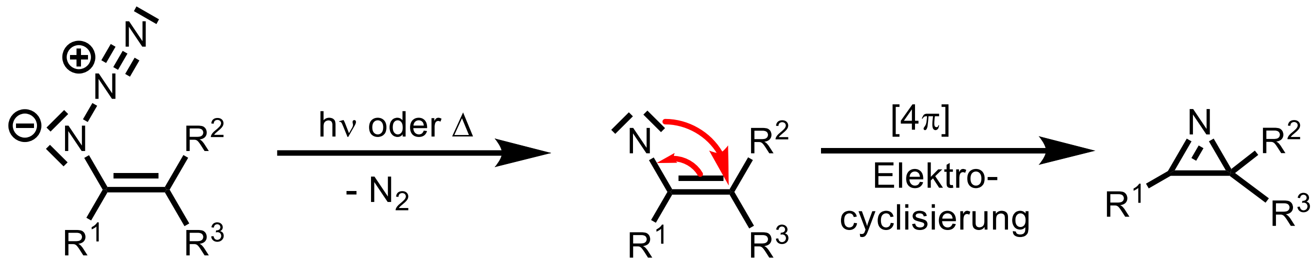
Azirinomycin
aus *Streptomyces aureus*

3.1.1.3. 2H-Azirin
Fortsetzung

b) Aus methylierten Dimethylhydrazonen

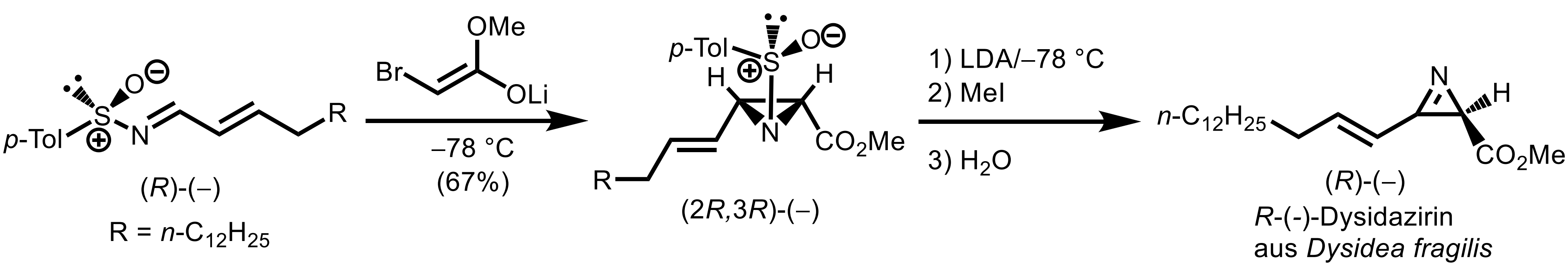


c) Thermo- oder Photolyse von Alkenylaziden



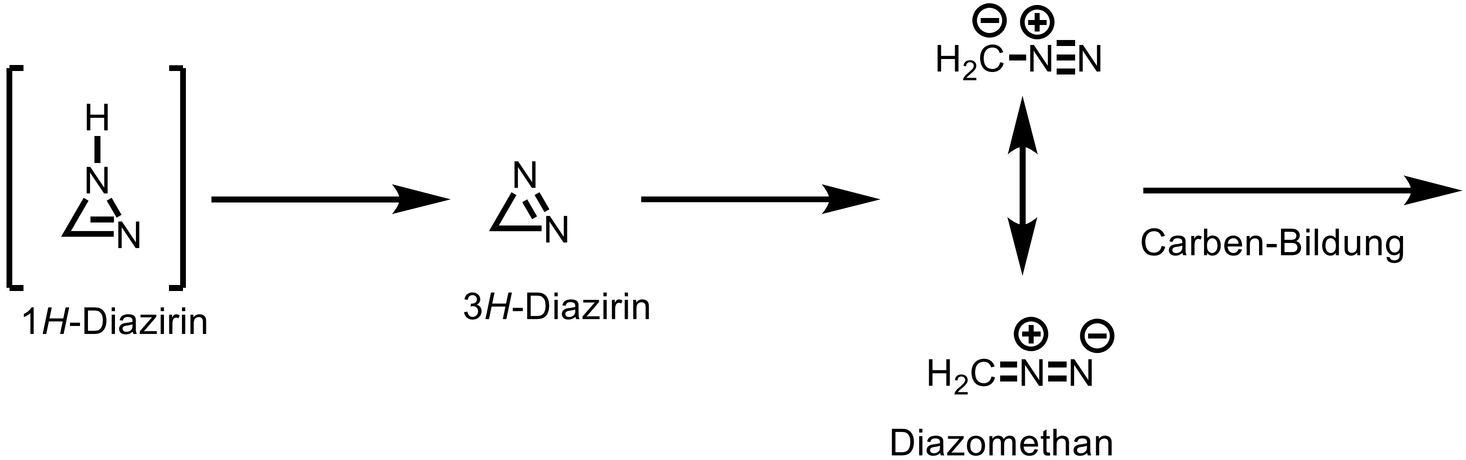
d) Eliminierung aus Aziridin

Asymmetrische Synthese, Cytotoxisches Antibiotikum (*R*)-(-)-Dysidazirine: F. A. Davis, G. V. Reddy, H. Liu, *J. Am.* Chem. *Soc.* **1995**, *117*, 3651.



3.1.1.4. 3*H*-Diazirin

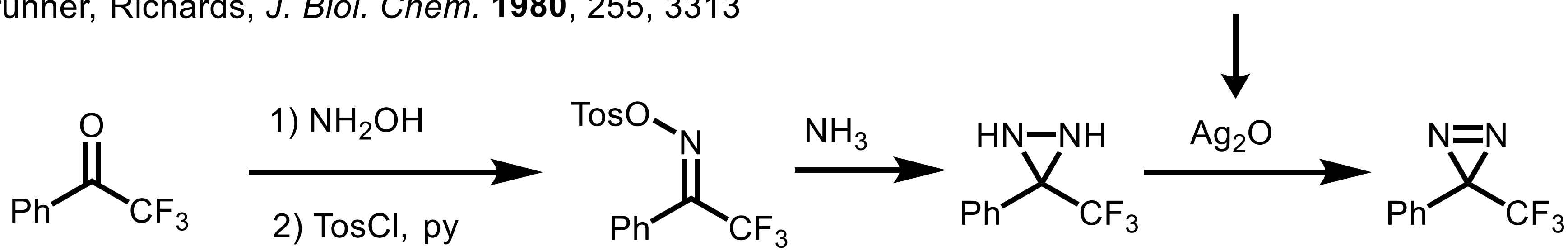
13



Synthese

Brunner, Richards, *J. Biol. Chem.* **1980**, 255, 3313

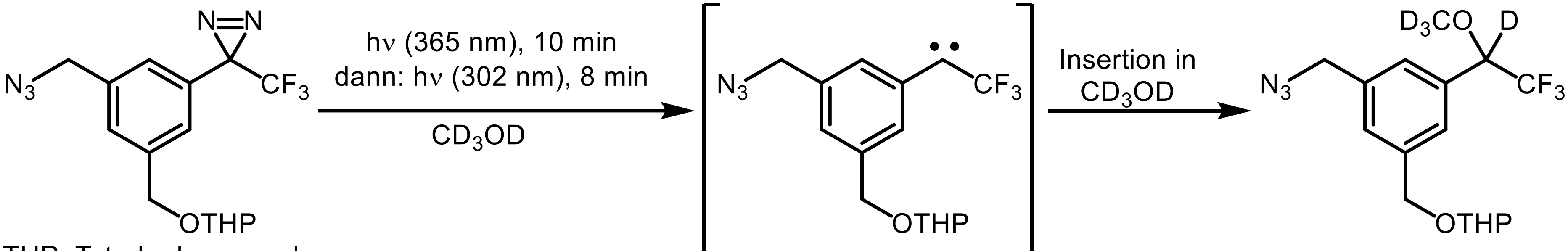
Paulsen 1960
Schmitz 1961



3.1.1.4. 3H-Diazirin Fortsetzung

3*H*-Diazirine sind Carbenquellen

3*H*-Diazirine in der Photoaffinitätsmarkierung: T. Hosoya et al., *Org. Biomol. Chem.* **2004**, *2*, 637.



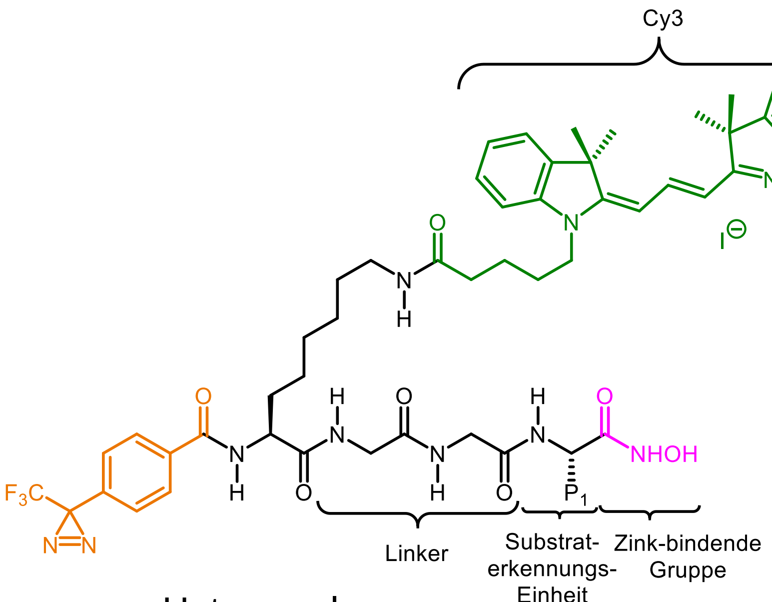
THP: Tetrahydropyranyl

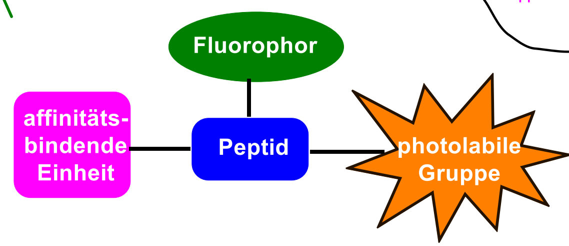
Azid bleibt erhalten!

>95% (^{19}F-NMR)

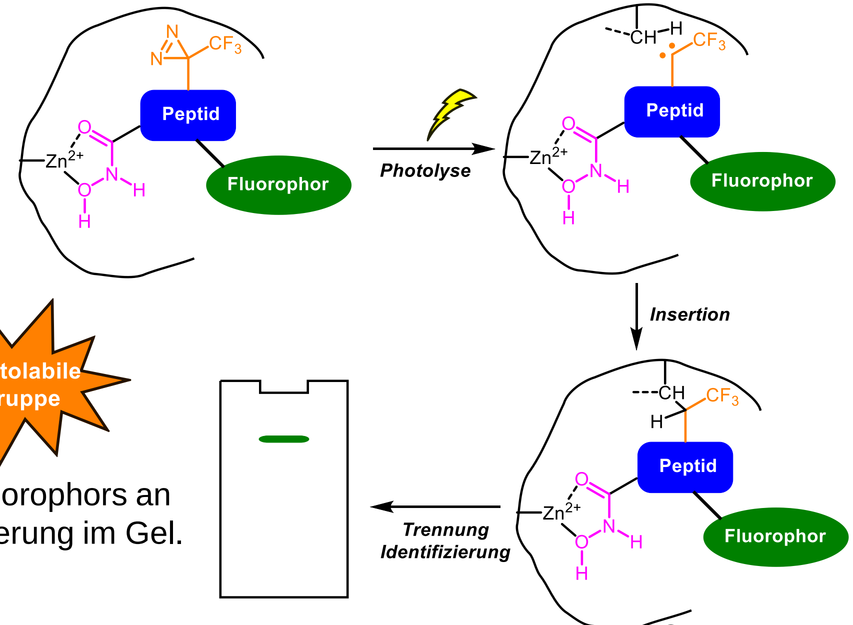
82% nach präparativer DC

Photoaffinitätsmarkierung von Proteinen (hier selektive Hydroxamat-Sonde für Metalloproteasen mit Zn^{2+} im aktiven Zentrum): E. W. S. Chan et al. *J. Am. Chem. Soc.* **2004**, *126*, 14435.





Kovalente Verknüpfung des Fluorophors an das Protein => leichte Identifizierung im Gel.



3.1.2. Gesättigte Dreiringheterocyclen

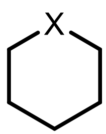
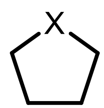
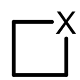
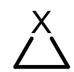
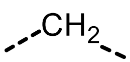
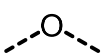
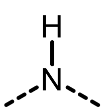
Exkurs: Ringspannung

"Baeyer-Spannung(senthalpie)" (1885): Abweichung der Verbrennungsenthalpie *pro Ringposition* vom Fall des Cyclohexans.

14

38.5	27.2	5.4	0	[kJ/mol]

Absolutwert der Verbrennungswärme pro CH_2-Gruppe bei Cyclohexan: 658.6 kJ/mol

Substituent X				
	0.2	6.4	25.7	27.3
	0.0	**6.2**	**26.5**	**27.5**
	1.2	5.7	24.9	26.4
	0.0	**5.4**	**24.7**	**26.3**
	1.1	6.1	25.4	27.0
	0.0	**5.8**	**25.2**	**26.7**

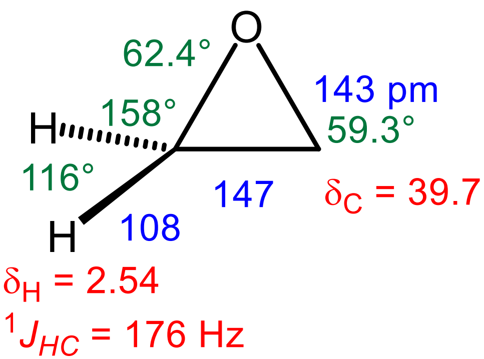
Spannungsenthalpien
in kcal/mol (1 kcal/mol = 4.184 kJ/mol)
(siehe *JOC* **2002**, 3488)

berechnete und **experimentell** bestimmte Werte

Ringspannung der 3- und 4-Ringe sehr ähnlich:
Heteroatom haben kaum einen Einfluss

Steckbrief von Oxiran, Aziridin und Thiiran

Oxiran

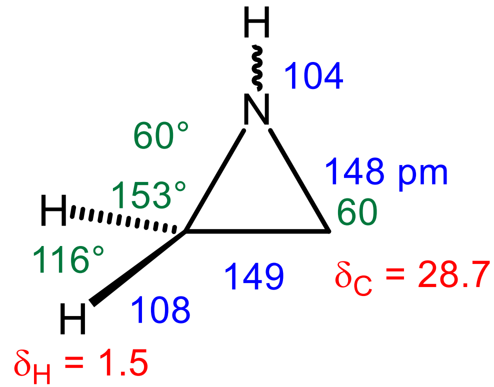


Spannungsenthalpie 114 kJ/mol

Sdp 11 °C; Produktion: 7 · 10^6 t/a

Ionisierungsenergie: 10.5 eV (Elektron aus nichtbindendem Elektronenpaar des O);
δ (^{1}H) = 2.54 ppm, δ (^{13}C) = 39.7 ppm;
Brönsted-Base, Lewis-Base

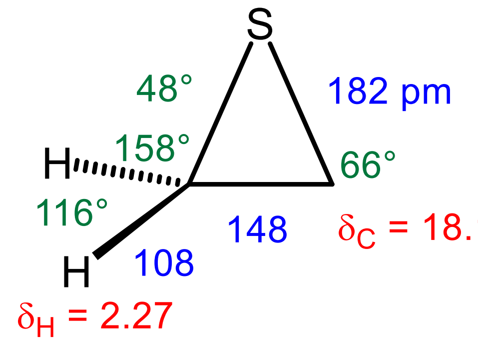
Aziridin



113 kJ/mol
Sdp 57 °C

Ionisierungsenergie: 9.8 eV
δ (^{1}H) = 1.5, 1.0 ppm, δ (^{13}C) = 28.7 ppm

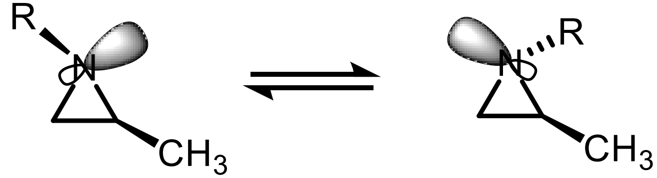
Thiiran



83 kJ/mol
Sdp 55 °C

Ionisierungsenergie: 9.05 eV
δ (^{1}H) = 2.27 ppm, δ (^{13}C) = 18.1 ppm
farblose, in Wasser kaum lösliche Flüssigkeit

15



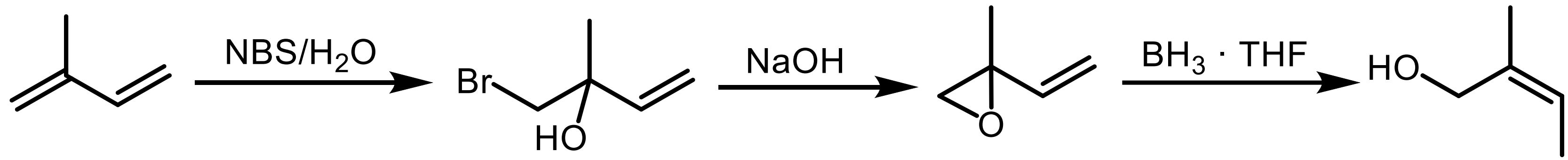
pyramidale Stickstoff-Inversion: R = Me: 70 kJ/mol; R = Cl: 112 kJ/mol

3.1.2.1. Oxiran

Synthesen

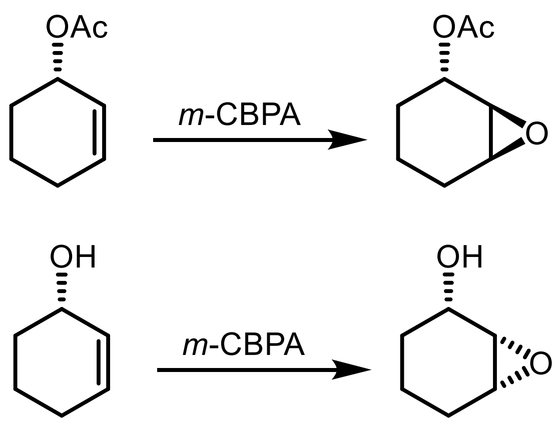
a) Aus Halohydrinen mit Basen (elektronenreiche Alkene reagieren rascher)

16



b) Aus Alkenen mit Persäuren (Prilezhaev-Reaktion)

17

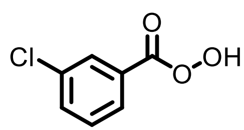


Reagenzkontrolle (von der leichter zugänglichen Seite)

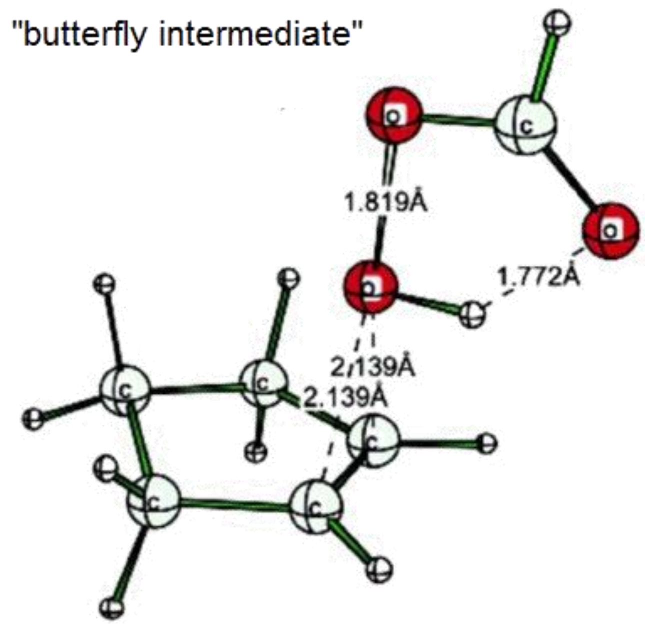
aber:

Substratkontrolle (H-Brückenbindung dirigiert auf dieselbe Seite)

m-CBPA: *meta*-chloro perbenzoic acid
m-Chlorperbenzoesäure



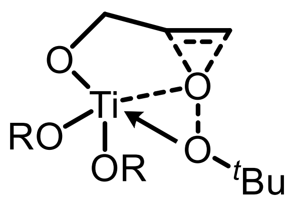
"butterfly intermediate"



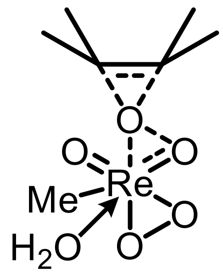
Berechnung des ÜZ: R. D. Bach, O. Dmitrenko, W. Adam, S. Schambony, *J. Am. Chem. Soc.* **2003**, *125*, 924.

Fortsetzung

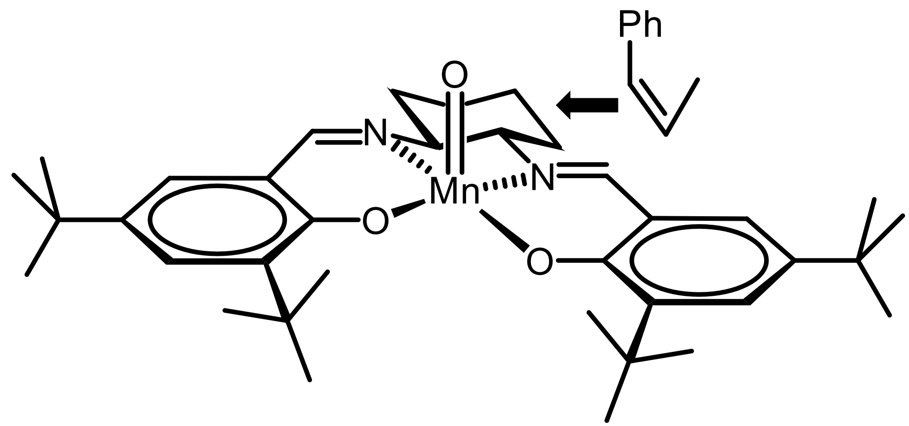
c) Übergangsmetallkatalysiert



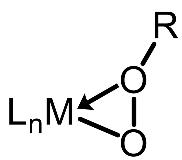
Peroxy-Komplex
Sharpless-Epoxidierung



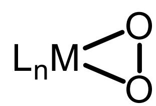
Peroxo-Komplex
Herrmann-Epoxidierung



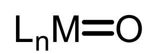
Oxo-Komplex
Jacobsen-Epoxidierung



Peroxy
M = Ti, V



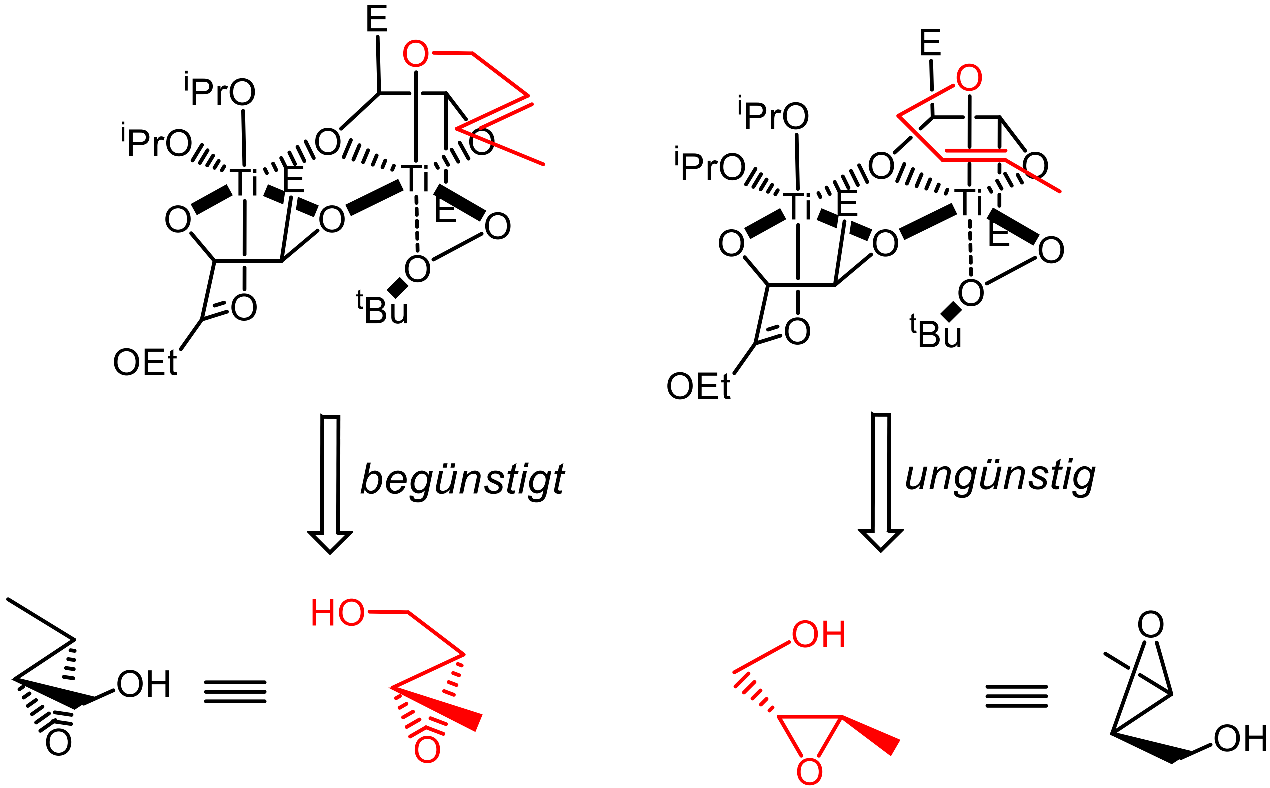
Peroxo
M = Re

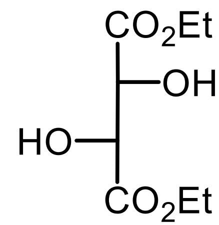


Oxo
M = Mn, Cr

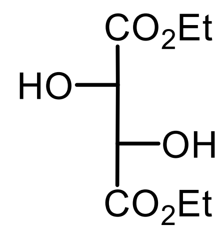
Fortsetzung

Enantioselektive Sharpless-Epoxidierung





L-DET (*R*,*R*-(+)-DET)



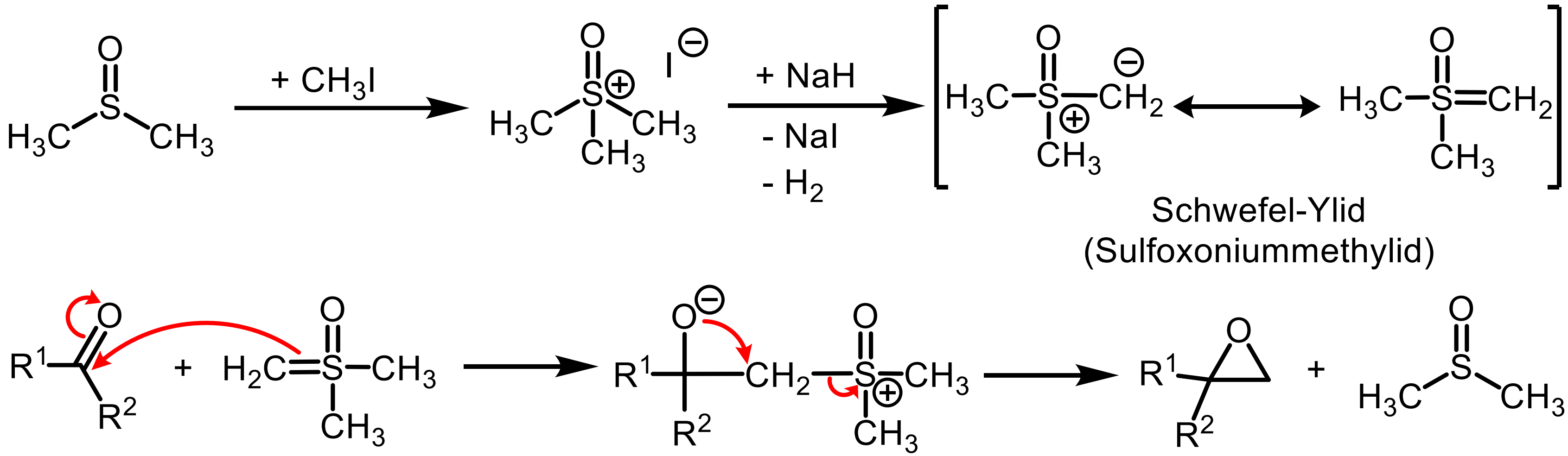
D-DET (*S*,*S*-(-)-DET)

18

3.1.2.1. Oxiran
Fortsetzung

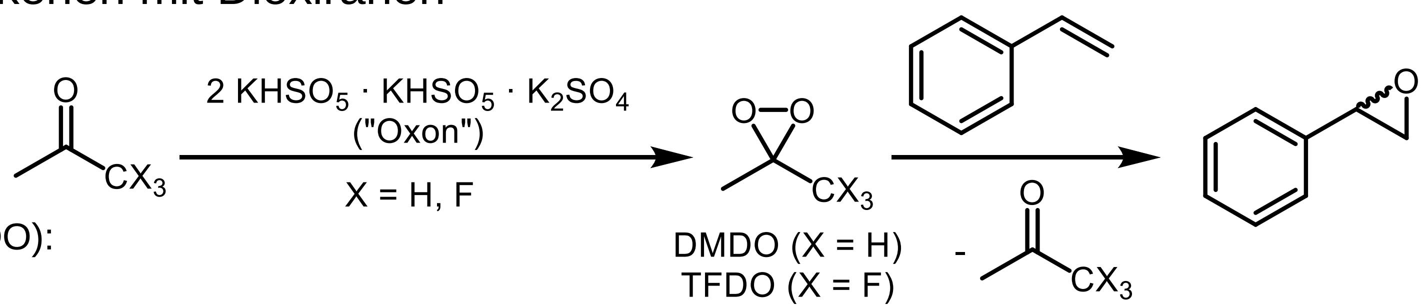
d) Corey-Synthese

19



e) Epoxidierung von Alkenen mit Dioxiranen

Dimethyldioxiran (DMDO):
Murray, *JACS* **1984**, 2462;
JOC **1985**, 2847.
Adam, *JOC* **1987**, 2800.
Trifluordimethyldioxiran (TFDO):
Curci, *JOC* **1988**, 3890.



Reaktivität gegenüber Alkenen:
TFDO (100000) >
DMDO (100) >
Peroxybenzoesäure (1)

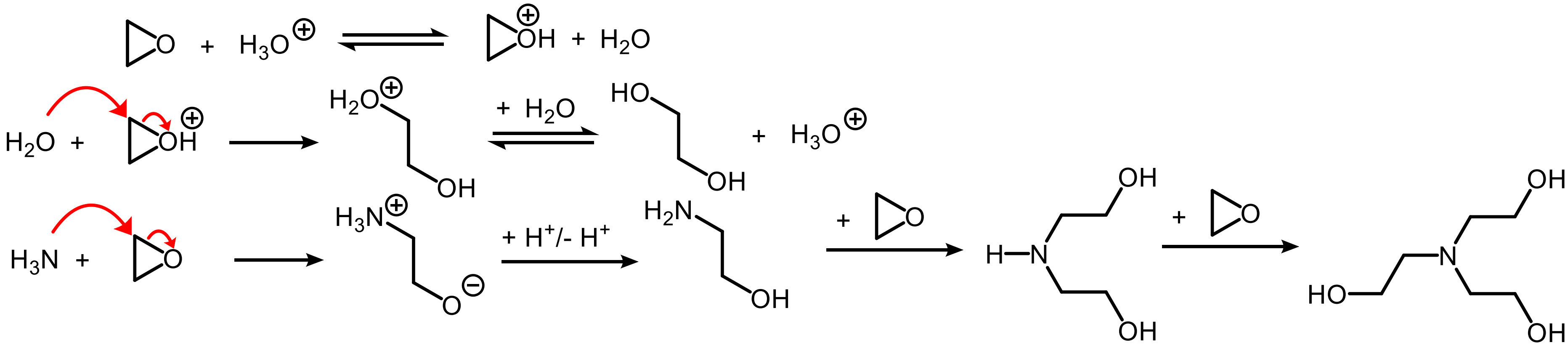
Ringspannung:
DMDO: ca. 75 kJ/mol
Dioxiran: ca. 105 kJ/mol
TFDO: ca. 110 kJ/mol)
Oxiran: ca. 115 kJ/mol

- Triebkraft: Bruch der O-O-Bindung
- O-Elektrophilie entscheidend für die RG
- Ringspannung ist irrelevant

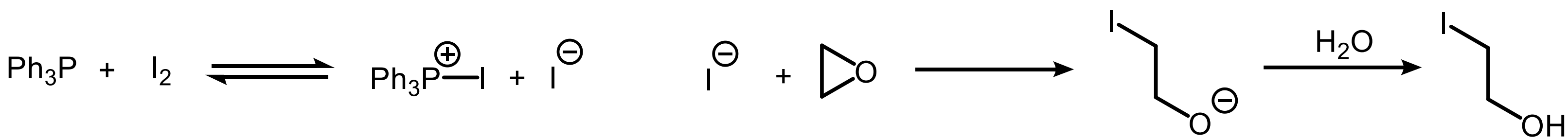
3.1.2.1. Oxiran
Fortsetzung
Reaktionen

20

Ringöffnungen durch Nucleophile

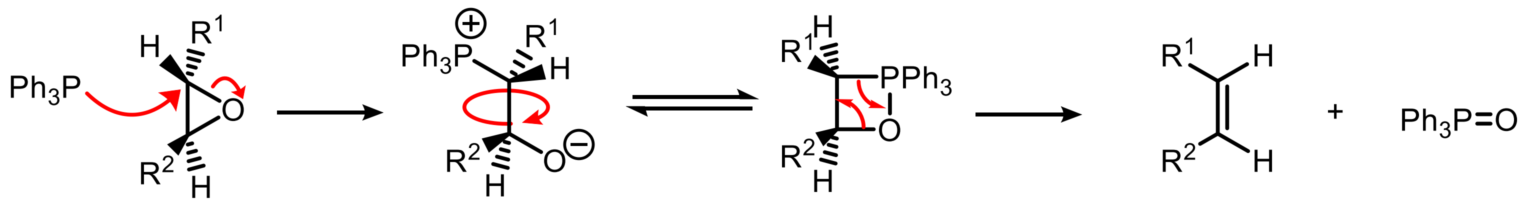


Ethanolamin   Diethanolamin   Triethanolamin



21

Deoxygenierung zu Olefinen

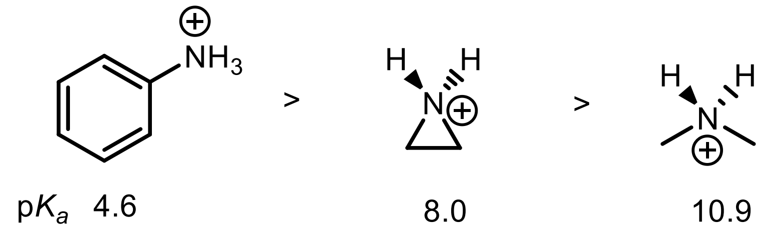


durch Epoxidierung
von *trans*-Olefin

3.1.2.2. Aziridin

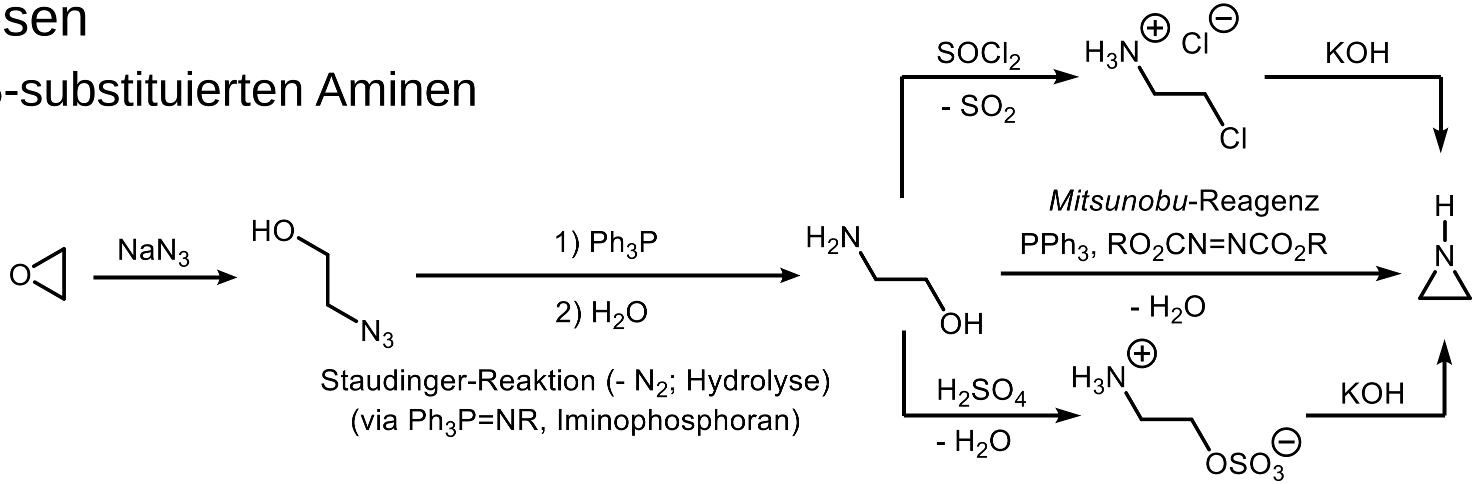
Basizität (Acidität der konjugierten Säure)

22

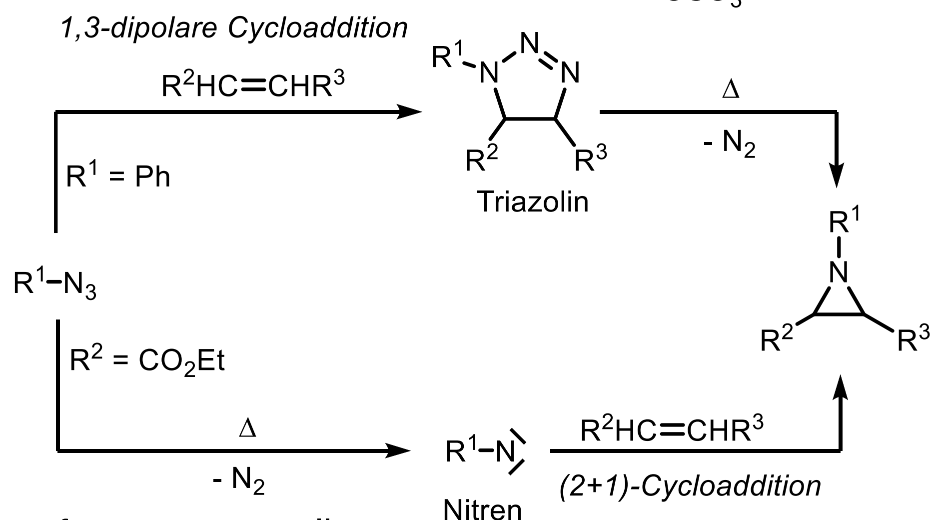


Synthesen

a) Aus β-substituierten Aminen

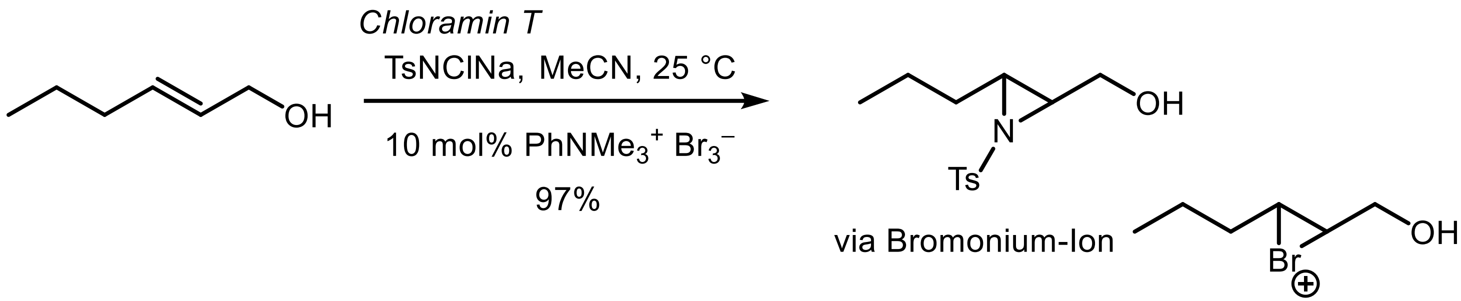


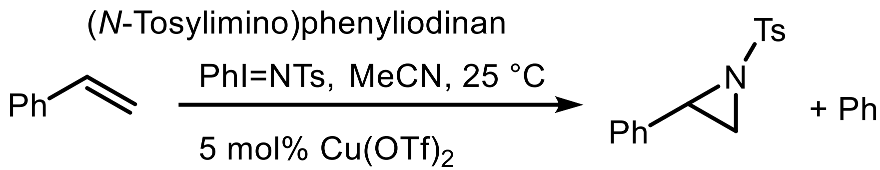
b) Aus Olefinen und Aziden



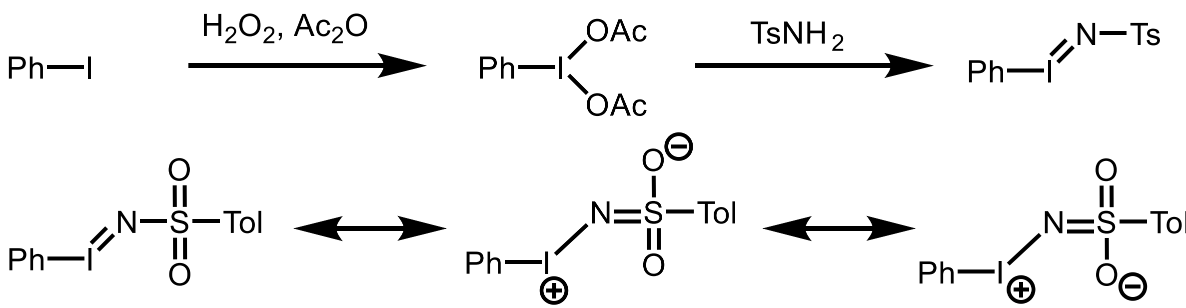
3.1.2.2. Aziridin
Fortsetzung

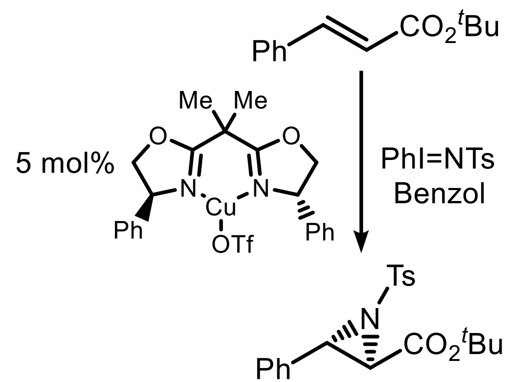
c) Aus alternativen Nitrenquellen





Evans et al., *JOC* **1991**, 6744; *JACS* **1994**, 2742; Dodd et al., *Synlett* **2003**, 1571.
Fe(III)-, Mn(III)-katalysiert: Mansuy et al. *JCSCC* **1984**, 1161.



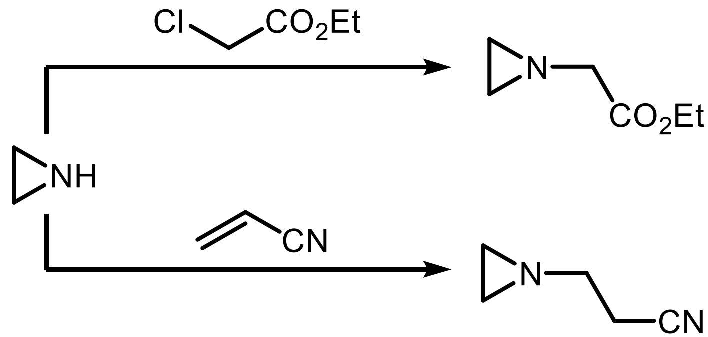


Bisoxazolin-Cu-Komplexe: Evans, *JACS* **1993**, 5328.
Salen-Cu-Komplexe: Jacobsen, *JACS* **1993**, 5326.

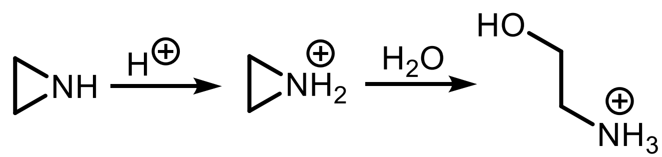
3.1.2.2. Aziridin
Fortsetzung

Reaktivität

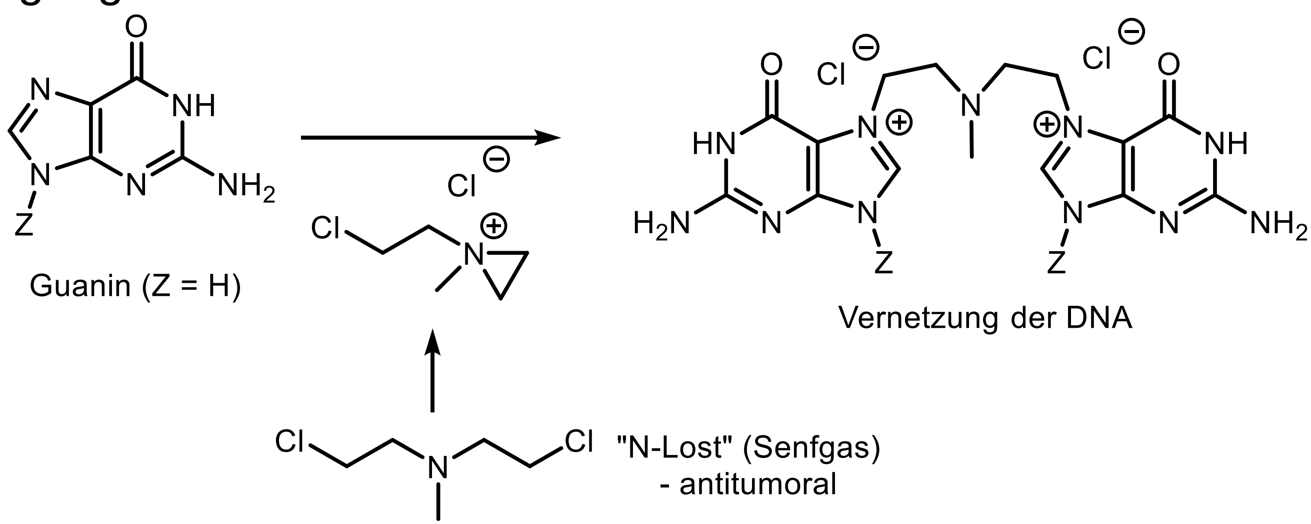
a) Gegenüber Elektrophilen



b) Elektrophilkatalysierte Ringöffnung durch Nucleophile

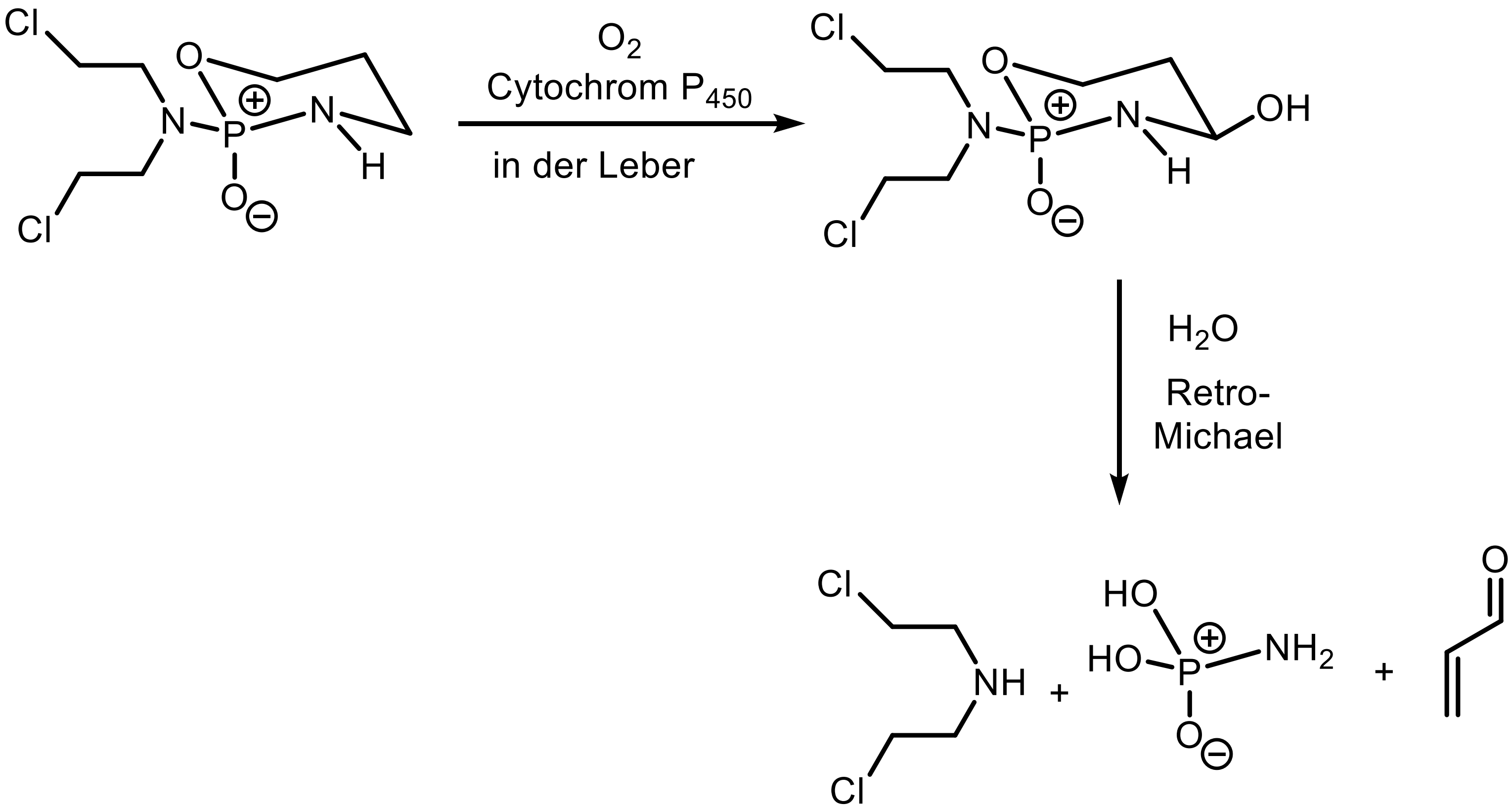


z. B. Schädigung von Nucleinsäuren



3.1.2.2. Aziridin
Fortsetzung

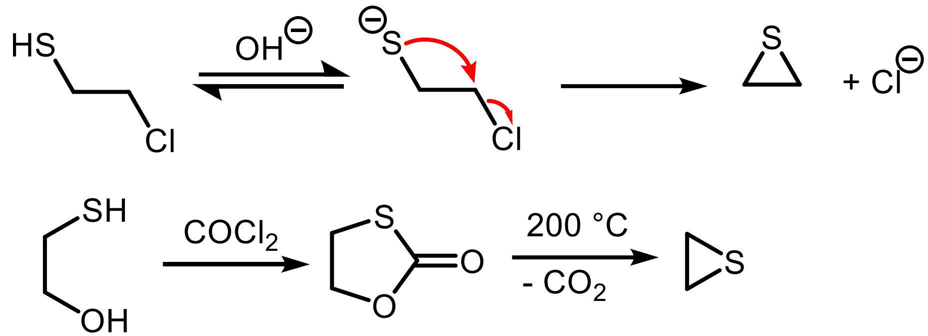
Therapeutikum Cyclophosphamid (maskierter N-Lost)



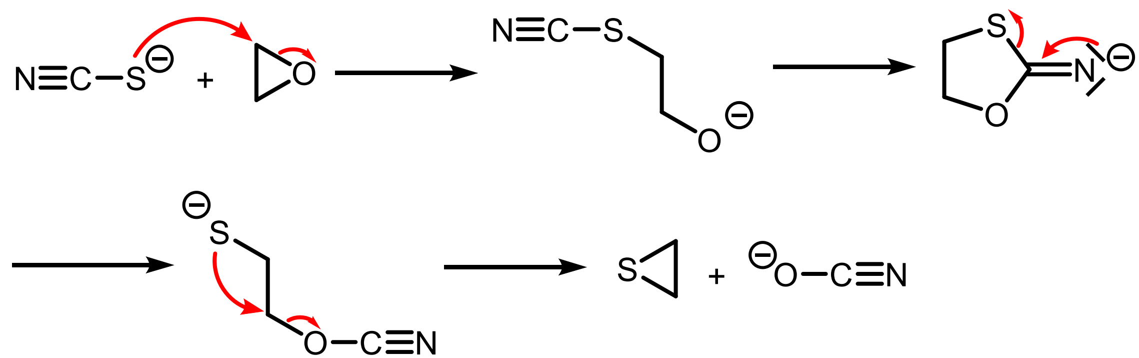
3.1.2.3. Thiiran

Synthesen

23 a) Cyclisierung von β-substituierten Thiolen



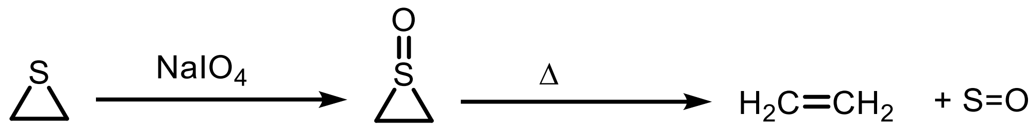
b) "Ringtransformation" von Oxiranen



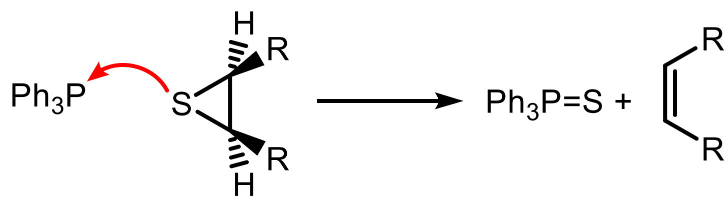
Fortsetzung

Reaktionen

a) Oxidation zu S-Oxiden

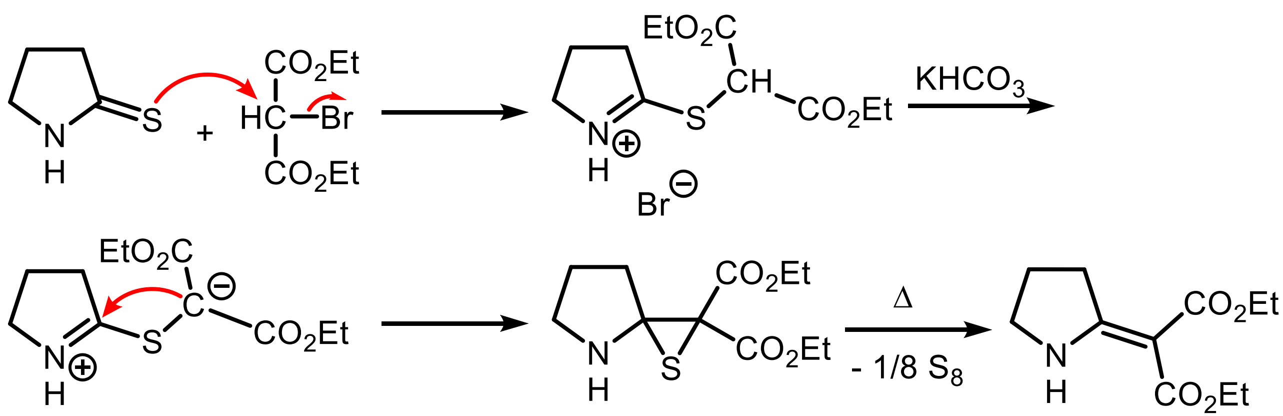


b) Desulfurierung zu Olefinen

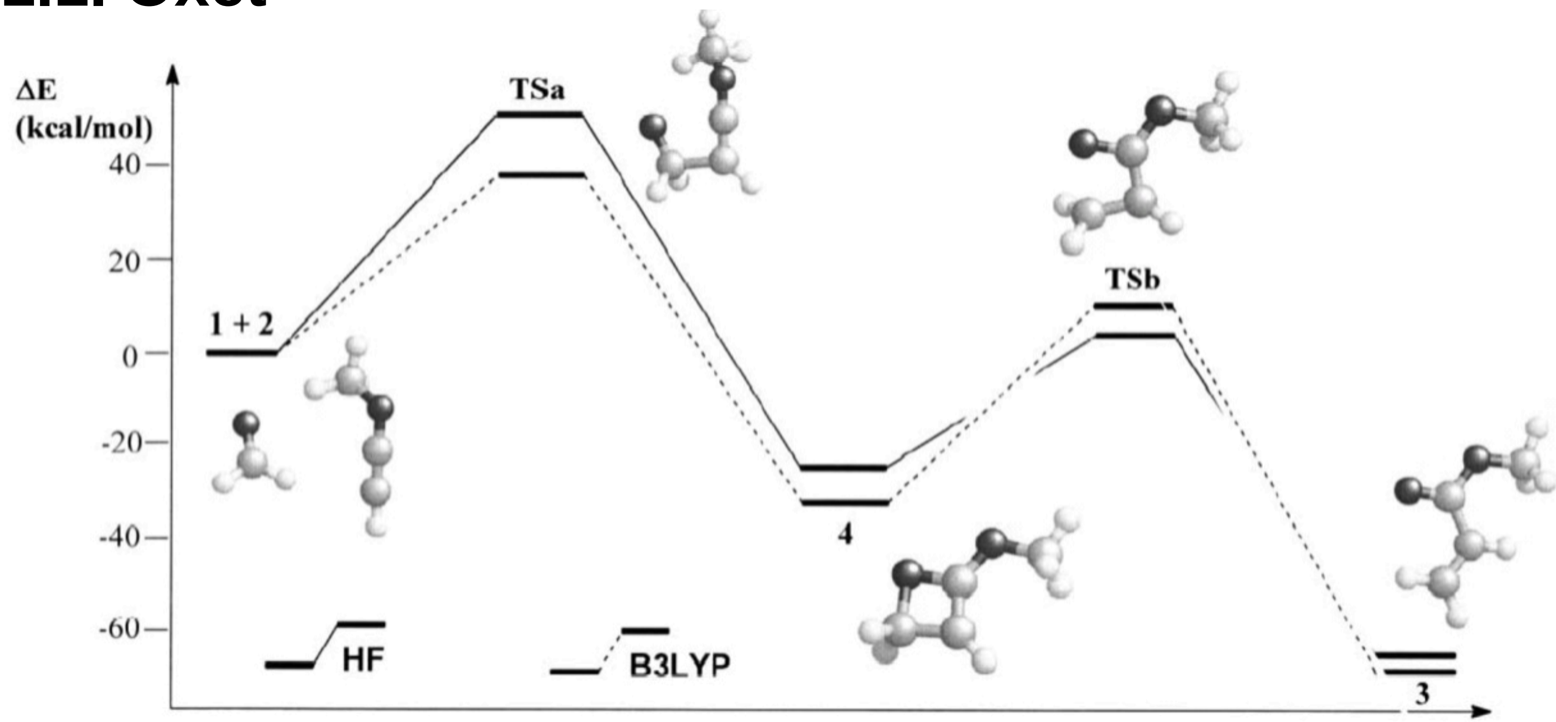


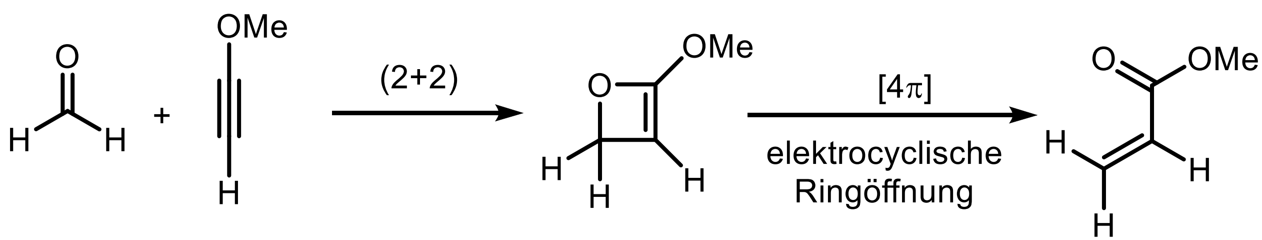
c) Sulfid-Kontraktion nach Eschenmoser

24



3.2. Vierring-Heterocyclen
3.2.1. Ungesättigte Vierringheterocyclen
3.2.1.1. Oxet

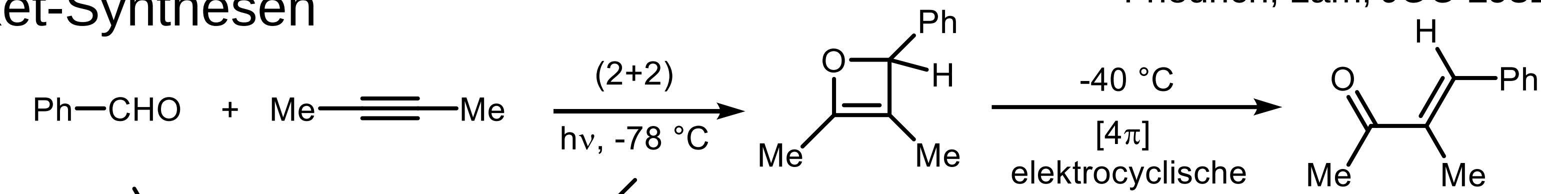


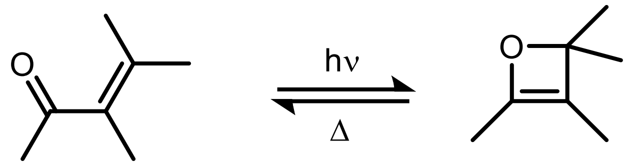


1 **2** **3** **4**

Oxet-Synthesen

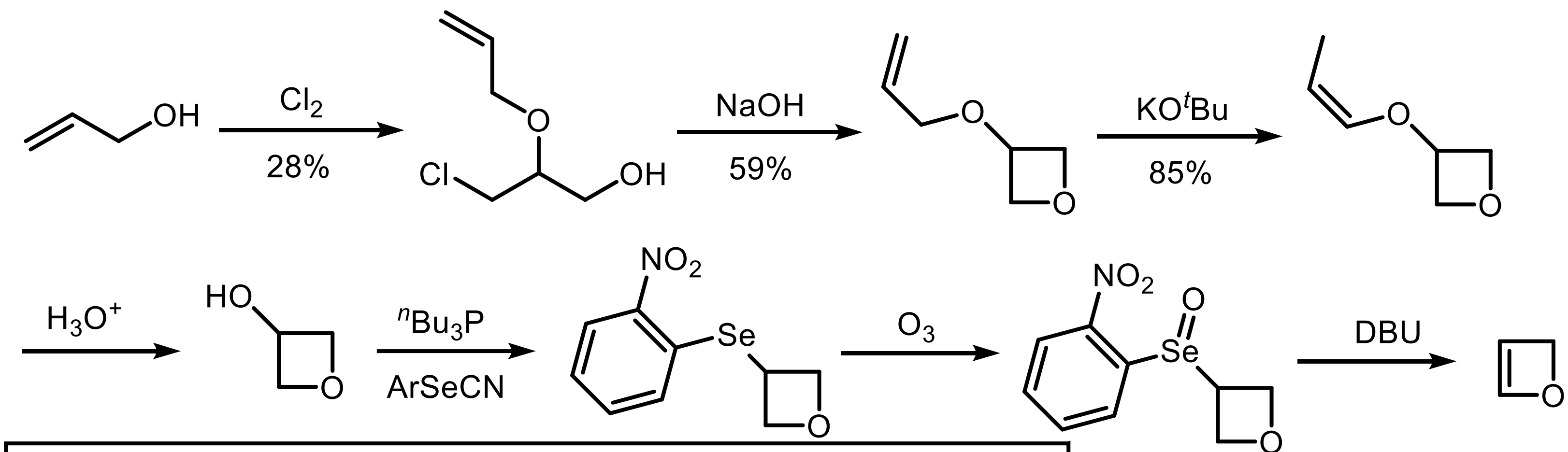
Friedrich, Lam, *JOC* **1981**, 306.

a) 

b) 

c) Stammverbindung

Grieco, *JOC* **1976**, 1485.



25

$$ArSeCN + {}^{n}Bu_3P \longrightarrow ArSe\overset{\oplus}{P}{}^{n}Bu_3\ CN^{\ominus}$$

$$ArSe\overset{\oplus}{P}{}^{n}Bu_3\ CN^{\ominus} + RCH_2OH \longrightarrow RCH_2O\overset{\oplus}{P}{}^{n}Bu_3 + HCN + ArSe^{\ominus}$$

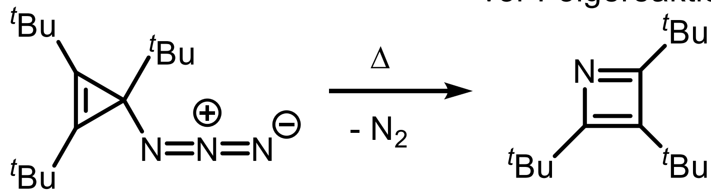
$$RCH_2O\overset{\oplus}{P}{}^{n}Bu_3 + ArSe^{\ominus} \longrightarrow RCH_2SeAr + O{=}P^{n}Bu_3$$

3.2.1.2. Azet

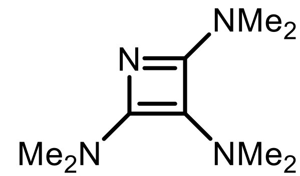
isoelektronisch zu Cyclobutadien (4π-Antiaromat); Stammverbindung unbekannt

sterische Abschirmung vor Folgereaktionen

elektronische Absenkung durch Donorsubstitution



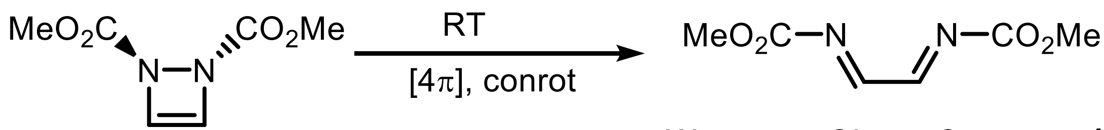
Smp 37 °C
Regitz (TU KL) 1986



Gompper (LMU) 1973

(2+2)-Cycloreversion thermisch verboten nach Woodward-Hoffmann-Regeln

3.2.1.3. 1,2-Dihydro-1,2-diazet

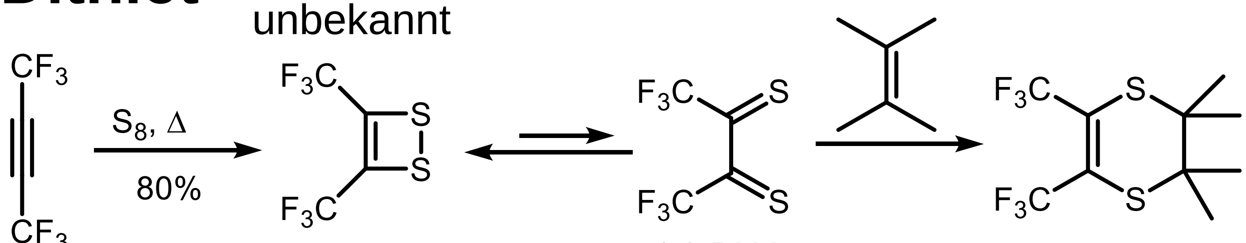


1,2-Dihydro-1,2-diazet
(nicht aromatisch, weil nicht planar!)

Warrener, *Chem. Commun.* **1972**, 818.

3.2.1.4. 1,2-Dithiet

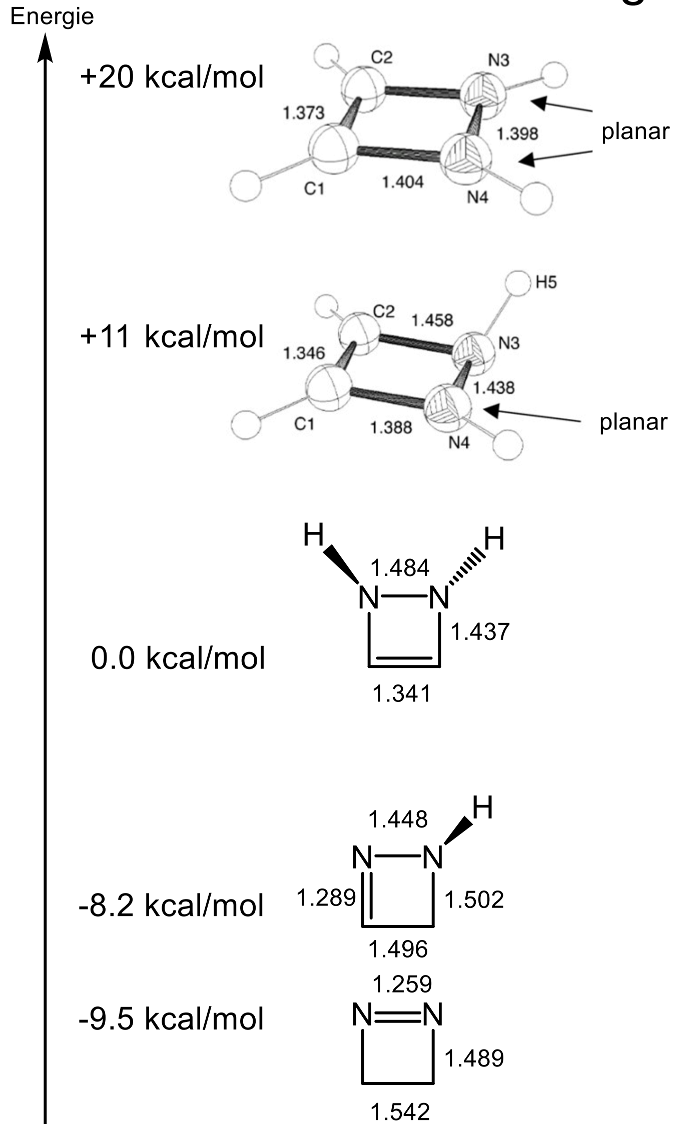
isoelektronisch zu Benzol (6π-Aromat), Stammverbindung trotzdem unbekannt



1,2-Dithiet

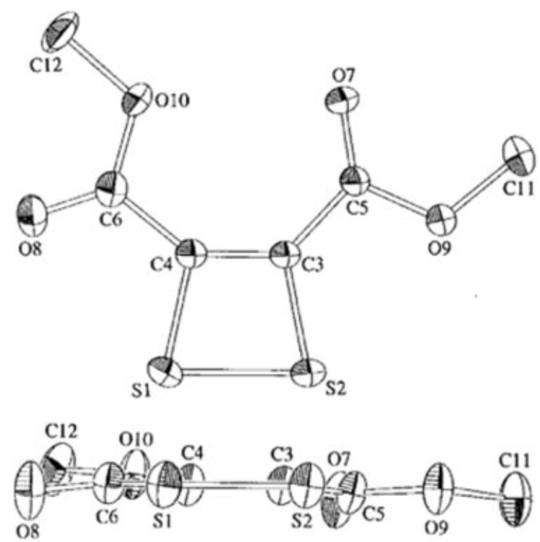
1,2-Dithion

Einschub: Sind 6π-Vierringe aromatisch?



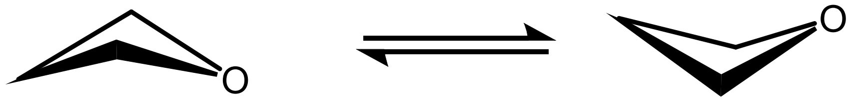
Breton, *JOC* **2002**, 6699; Budzelaar, *JACS* **1987**, 6290
Theoretische Studie auf DFT-Niveau zur Stabilität der diversen Isomere (RB3LYP/6-311+G(2d,p)
⇒ 1,2-Dihydro-1,2-diazet ist nicht aromatisch

1,2-Dithiet ist im Kristall planar.
(Kamigata, *JOC* **1998**, 6192.)



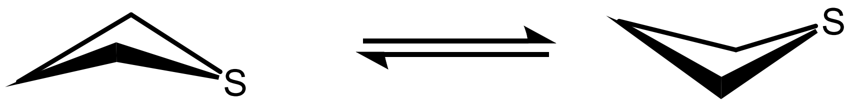
3.2.2. Gesättigte Vierringheterocyclen

Oxetan, Thietan, Azetidin



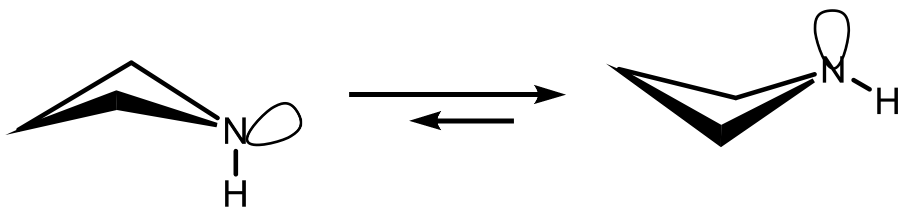
Oxetan (Sdp 48 °C)

Ringspannung: 106.3 $kJmol^{-1}$
Inversionsbarriere 0.18 kJ/mol ⇒ geringer als die Schwingungsenergie ⇒ rasche Inversion



Thietan (Sdp 94 °C)

Inversionsbarriere 3.28 kJ/mol ⇒ oberhalb der niedrigsten 4 Schwingungsniveaus ⇒ langsamere Inversion ⇒ Ring ist nicht planar



Azetidin (Sdp 62 °C)

Inversionsbarriere 5.5 kJ/mol
(vgl. Cyclobutan: 6.2 kJ/mol)

Verringerung der Pitzer-Spannung durch Faltung (“puckering“). Spannungsenthalpie ist nur 3-7 kJ/mol geringer als die der Dreiring-Heterocyclen.

Gesättigte Vierringheterocyclen sind wie die Dreiring-Heterocyclen durch S_Ni (Cyclisierung) der 3-Halogenverbindungen herstellbar.